

GAURAV KUMAR

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Education

MAY 2019	B. Tech. MECHANICAL ENGINEERING IIT Kanpur	7.6/10.0
APRIL 2015	Class XII (CBSE) SECONDARY DELHI PUBLIC SCHOOL,GAYA	94.8%

Technical Skills

Programming Skills:	Rust: 6y, C++: 3y, Python: 2y, Go: 3y, C: 2y
AI Skills:	RAG, Prompt Engineering, Vector Databases, Embeddings, MCP
Container Management:	Docker, Kubernetes, OpenShift, Helm, Operator, Docker Compose
Cloud:	Azure, AWS, GCP, IBM Cloud
Database:	Cassandra, CockroachDB, Qdrant, Redis
Debugging:	gdb, strace
Automation/Build:	Jenkins
Observability:	Grafana, Splunk

Work Experience

Lead Cloud Development Engineer at Citrix

July'25-Ongoing

- Enhanced Edge Service security by adding Web Application Firewall (WAF) and rate-limiting controls.
- Extended EDT NAK statistics to include range support.
- Built a cgo wrapper for the EDT C library to enable safe and idiomatic Go APIs
- Added EDT support in the CGS tool to enable secure, low-latency communication over UDP for virtual apps and desktops.

Senior Software Engineer at Fortanix Inc.

June'19-June'25

- ArmetAI Team**
 - Contributed to the design and development of a confidential Retrieval-Augmented Generation (RAG) pipeline.
 - Implemented core backend services and integrated all microservices.
 - Designed and implemented attestation mechanisms for a new AMD Confidential VM platform with GPU support, enhancing security and verification processes.
 - Architected and implemented security hardening for the Qdrant vector database.
 - Added protections against prompt injection, jailbreak attacks, toxicity, PII exposure, and hallucination in the RAG pipeline.
 - Packaged and configured Armet AI services for deployment in Customer-Managed VPCs.
- Confidential Computing Manager (CCM) Team**
 - Conceived and implemented a confidential data clean room.
 - Spearheaded the Multi-Party Support feature, enabling collaborative workflow creation and execution across multiple accounts.
 - Developed a microservice for workflow monitoring, enabling efficient management of workflows.
 - Played a key role in the launch of the CCM SaaS product, a revolutionary platform for confidential computing (<https://ccm.fortanix.com>).
 - Architected and implemented core back-end functionalities, including the development of initial REST APIs, metering, and audit-logging microservices.
 - Delivered advanced features such as DCAP attestation support, multiple container registry configuration per account, Node Agent packaging, and an Operator for the CCM Node Agent.
 - Enabled DCAP attestation support during secure backend cluster join, enhancing trust verification in distributed environments.
 - Migrated CCM services to run on Ubuntu 24.04.
- Data Security Manager**
 - Designed and developed a k8s plugin to encrypt secrets and ConfigMaps stored in etcd using DSM.
 - Implemented a plugin to securely inject secrets directly into Kubernetes workloads.